German International University
Faculty of Engineering, Department of manufacturing

Applying Lean Manufacturing for Improving the Performance

Bachelor Thesis

Author: Abdelraheem Zekry
Supervisors: Dr.-Ing. Amr Nounou
Submission Date: 18 January, 2024

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This is to certify that:

(i) the thesis comprises only my original work toward the Bachelor Degree

(ii) due acknowledgement has been made in the text to all other material used

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Abdelraheem Zekry
18 January,2024

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I would like to express my sincere gratitude to all those who contributed to the completion of this report. Their support and guidance have been invaluable throughout this endeavor. My supervisor, Dr.-Ing. Amr Nounou, has my sincere gratitude for his knowledge, guidance, patience and perceptive criticism, which have been instrumental in forming the format and content of this report. Your advice has been extremely helpful in navigating the subject's intricacies. Special thanks goes to Eng. Kareem Salah and all of Metrotec family for their assistance and guidance in applying the project in Metrotec facility, by providing the tools needed for the project, time to evaluate the modifications, monitoring and guiding the project, enhancing the overall quality of this report. Your dedication and attention to detail have been truly commendable.Finally, I would like to thank the professional and academic community whose efforts made this research possible. The extensive body of knowledge and insights from previous research and conversations has greatly influenced the context and in-depth analysis of this report. Thanks for all of support, encouragement and contributions.

Abstract

Lean manufacturing is a philosophy that was implemented by Toyota. Undoubtedly the lean philosophy became a widely adopted philosophy in various industries, regardless of the size of the facility or industry. This project will discuss problems affecting productivity in the manufacturing of water meters in Metrotec company, which is the top manufacturer of water meters in Egypt. The literature review will explore the lean tools that are used in different industries which result in eliminating wastes, in addition to case studies that applied lean manufacturing and its effect on the facility. A methodology was conducted by implementing a flowchart, explaining explicitly the approach used to implement the lean manufacturing in this project. As the aim of the project is to improve the productivity using lean manufacturing tools, focusing on the machining process of the water meter body in Metrotec. The implementation section identifies the wastes in the machining process, in addition to planning improvements to solve problems in the process. Finally, evaluating the planned improvements through several runs and discussing the results of the project on the facility.

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Chapter 1

Introduction

1.1 Background

Over the past century, there has been a major revolution in utilizing resources efficiently in industry. Lean manufacturing (LM) is one of the main concepts that evolved in the mid-1900s was lean manufacturing[19]. Lean manufacturing is a technique that concentrates on reducing waste in production systems while increasing productivity. In order to execute Lean Manufacturing, businesses must use approaches that encompass critical variables such as integrated management systems, strategy and culture, and operations aimed at improving performance. Lean manufacturing originated in the automobile sector and has now spread to engineering and assembly businesses[9]. It was largely created by Toyota founder Sakichi Toyoda, his son Kiichiro Toyoda, and Toyota chief engineer Taiichi Ohno. Furthermore, lean systems have three main elements, which are continuous improvement, waste reduction and demand-driven. Therefore, the ultimate goal is to achieve a smooth, rapid flow of materials through the system to match customer demand[7]. The adoption of the VSM tool resulted in lower levels of inventory and rework. In addition to the advantages of the production flow, it was said that the employment of (LM) tools optimizes gains in the environmental and social aspects of the manufacturing process[8]. Companies use Lean Manufacturing to maintain their competitiveness by increasing the productivity of their manufacturing systems and enhancing product quality. As a result, most academic and manufacturing researchers are drawn to lean manufacturing approaches. Each activity must be evaluated before implementation, from design to dispatch; additionally, each individual in the organization pyramid will be responsible in some way, with the authority to make decisions at his or her own level in real time [14].

No industry is exempt from the potential advantages of lean manufacturing, including the water meter business. Water meter firms have recently come under intense pressure from changing market dynamics, heightened competition, and changing consumer expectations to innovate their processes, shorten lead times, and boost flexibility while maintaining or even lowering prices. The concepts of lean manufacturing provide a methodical

way to tackle these problems successfully. In the setting of a water meter plant, this project study aims to apply the Lean Manufacturing principles and eliminating wastes, by carrying out a thorough case study in Metrotec a water meter facility. Metrotec is the origin of water meter industry in Egypt and considered to be the top facility in the Egyptian market and one of the most productive in the middle east. Metrotec is certified ISO 9001:2015 since 2018. Metrotec is the sole agent Of Zenner International The German Leader Of Water Meter Manufacturing, with over 100 years of experience.

1.2 Project charter

METROTEC EGYPT

Project Charter

Project Title	Applying Lean Manufacturing for improving the Performance		
Project Leader	Eng.Kareem Salah El Deen	Site	Metrotec Egypt for Water Meter
Objective	Increasing the productivity in the machining process.		

Start Date:	Target Completion Date
1/10/2023	31/12/2023

Team Members	Function
Eng.M.Elhawy	Production engineer
Mr.Magdi Taha	Machining Head
Mr.Saad Zaghloul	Machining Supervisor
Mr.Saeed Elwan	Machining Supervisor
Mr.Mahmoud Bayoumi	Cutting Tool maintenance head

Problem Description and Scope
Machining of water meter body passes through 7 stages, the first stage has the least productivity which leads to delay in the subsequent stages. Which affect negatively the delivery time of the water meter body to the assembly department and by the end to the customer.

Business Impact
Increase of productivity of the water meter body by 8.3% , decrease of body cost by %

Project Metrics	Current State (Baseline)	Objective and Stretch Target	Stakeholders	Signature	Dated
Cost variance (CV)		Decrease cost by % -0.94% From Machining Cost Per body 23/4	Sponsor:		16-1-2024
			Team Leader:		16-1-2024
			Finance Rep.:		

Figure 1.1: Project charter

This Charter is a prove of doing this project in Metrotec facility and dignifies the efforts done in the project by visiting the facility 2 or 3 days per week for 4 months to work on this project.

1.3 Problem statement

The project aims to increase the productivity of the mechanical water meter (Zenner), by studying the wastes that affect the production process, using lean philosophy. The

production of the water meter consists of 20 processes. The project will focus on the bottleneck in the machining part that includes 5 processes by decreasing its C/T and achieving the targeted C/T from the assembly department as the C/T of the machining processes is higher than the assembly. Wastes are the main problem that is hindering the production of the water meter. These problems will be discussed thoroughly and their solutions in the upcoming sections.

Chapter 2

Literature review

2.1 Introduction

While there has been much research on applying lean manufacturing in various industries, few researches were found[2]applying lean manufacturing in water meter facilities into consideration. This literature review provides an overview of key concepts, principles, and research findings related to lean manufacturing. Lean manufacturing, derived from the Toyota Production System (TPS), has evolved into a widely accepted and applied philosophy for operational excellence across industries.

2.2 Lean philosophy

It is a philosophy that can be implemented in any industry, regardless of how large the production facility is. To implement this philosophy, there are five principles that should be followed. These principles are first, identifying the value needed by the customer. second, value stream mapping. This includes all the steps and processes involved in taking a specific product from raw materials and delivering the final product to the customer. Third, smooth flow of the product or service. Fourth, pull systems which means produce the product as much as needed by the customer. Fifth, strive for perfection [6]. In addition to the elimination of waste and the generation of flow, As a result, lean manufacturing provides numerous advantages, such as reduced inventory, financial savings, reduced rework, reduced inventory, and reduced lead time [12].

2.3 Wastes

To apply this lean philosophy, several tools are used to eliminate waste. Waste is any activity that doesn't add value to the customer. There are eight main types of waste.

Overproduction is producing more than needed in a product, regardless of the demand for the product. This leads to high inventory costs. Workers or equipment are waiting to do an operation during working hours due to an inefficient flow of materials or layout or uneven production speed in every station. (Transportation) includes the movement of tools, materials, equipment, and products without adding value to the customer. (Inventory) Although high inventory is an asset, it offers a discount for bulk purchases and maintains a steady work flow in high demand. However, it hides various problems such as bottlenecks, poor timing, defects, and work-in-progress products. (Over processing): adding a particular process that doesn't add value to the product. Particularly over functioning products. (Motion) includes unnecessary movement between departments. This could be due to inefficient flow between departments. (defects) errors during the process of production, which leads to rework or scrap. Which results in additional costs with no added value. (Unused talents): underestimation of workers talents and capabilities, or even placing workers in places they don't fit in[12] [10].

2.4 Theory of constraints(TOC)

The theory of constraints is used to identify the limiting factor for achieving a goal in the manufacturing system and improving that constraint until it is no longer limiting factor. The constraint is referred as a bottleneck. The TOC has 5 steps to eliminate the constraints. First, identifying the current constraint. Second, Exploit: make quick improvements to the constraints with existing resources. Third, Subordinate: Review all other activities to ensure they are aligned and support the needs of constraint. Fourth, Elevate: if the constraint still exists, consider further actions to be taken. Fifth, Repeat: make a sustainable cycle of improvements. If one constraint is resolved then move to another constraint[16].

2.5 Lean tools

2.5.1 Kanban

Kanban is a Japanese word which literally means a flag or sign. It is a Japanese manufacturing system in which is controlled by an instruction card issued along the production line. It allows for just-in-time manufacturing by meeting the demand for work when needed, eliminating inventory and waste. Types of Kanban are Visual signaling: Kanban communicates through visual indicators such as cards, bins, or electrical signals. These signals specify when production should begin, what should be produced, and how much should be produced. Pull System: pull system means that the production begins in response to actual demand from downstream processes. Therefore, overproduction and excess inventories are reduced. Limit Work in Progress: Kanban aims to decrease the number of WIP items, reducing resource overload and ensure a consistent work flow. The

following steps are used to implement a Kanban system. First, Visualize the Workflow by Identifying the phases in the workflow. Second, design Kanban Cards or signals to be used and associate it with a specific task. Third, define WIP boundaries to avoid overproduction and maintain a consistent flow. Fourth, continuously monitoring of the Kanban system to optimize the manufacturing processes. Finally, the Kanban system has various benefits, such as reduced excess inventory by manufacturing only what is needed when it is needed, better efficiency as Kanban encourages a smooth production processes, short lead times and more efficient resource utilization. Increasing flexibility, the pull system enables a fast response to changes in demand. Increasing quality as fewer products in development, there is a greater impact on product quality. [6][18] [**?**]

2.5.2 Just In Time

Just in time is a manufacturing philosophy that evolved in the 1950s by Toyota Production System (TPS). It aims to produce what the customer wants, when the customer needs it, and in what quantity. So, JIT is a pull system based on the demand of the customer. It has multiple advantages, including reducing lead time, inventory costs, and planning complexity. In addition to highlighting problems quicker, improving quality and productivity Finally, JIT has had a significant impact on several industries, including manufacturing, healthcare, and service. It is attributed to lowering costs, improving quality, reducing lead times, and enhancing customer satisfaction.gupta2013literature,[14],[19]

2.5.3 5S Methodology

This is a methodology developed by Toyota in the 1950s. The benefits of implementing 5S are creating a more effective work environment, reducing waste time from unnecessary motion and creating space. This is applied using five main principles (Sort) eliminate unneeded items and sort necessary and unnecessary items in the workspace. (Straighten) Organize what remains after sorting, using floor marks and signals. (Shine) apply routine maintenance, cleaning, and inspecting tools regularly. (Standardize) make a standard for the 5S by implementing schedules and checklists. Sustain: assures that 5S is applied on an ongoing basis. [10]

2.5.4 Poka Yoke (Mistake proofing)

It is used When a process step has been discovered where human error can result in errors or defects, particularly in operations that rely on the worker's attention, talent, or expertise, In a service process, the customer's fault affects the output. When output (or, in the case of service procedures, the client) is transferred to another worker during a process hand-off, When a tiny mistake early in the process generates huge issues later in the process, When the repercussions of a mistake are costly or dangerous. Poka yoke comes in two varieties: Shutdown: These are the most effective error-proofing tools. A machine, for example, will not start if a workpiece is wrongly set. Warning A buzzer or light alerts us to the presence of something strange. palange2021lean,[1]

2.5.5 Kaizen (continuous improvement)

Is a strategy that synchronizes all of the employees to achieve incremental improvement steps, regardless of how big the improvement is. it aims. entails the continuous pursuit of minor, incremental changes to processes, products, and systems in order to improve quality, efficiency, and overall performance. Kaizen encourages all employees, regardless of their organizational roles, to participate in the improvement process. *Continuous Improvement:* The pursuit of continual improvement in all parts of a business, from procedures and products to staff skills and knowledge, is the basic principle of Kaizen. 2. Standardization: The need for standardized work procedures as a foundation for development is emphasized by Kaizen. 3. Involvement of Employees: It encourages all employees, from the shop floor to the executive level, to actively participate in recognizing problems and coming up with solutions for improvement. Kaizen has a significant impact on company culture, instilling in employees a sense of responsibility, teamwork, and a desire for excellence. It leads to increased operational efficiency, superior product quality, less waste, and higher customer satisfaction. [19]

2.5.6 VSM

Value Stream Mapping is one of the most important lean tools for identifying the bottlenecks and productivity in a facility. VSM analyze value-added and non-value-added activities. Moreover, using VSM as a visual tool to help detect hidden waste and waste sources. [17]

2.6 Lean case studies

In the case study [17]applying VSM in a manufacturing process in a Malaysian automotive facility. The use of the VSM improved the approach to lean production initiatives by revealing visible and hidden waste that impacted on the front disc (D45T part) production productivity. A considerable portion of the time products spend in the manufacturing system is frequently wasted waiting and not adding value. Quantitative research revealed that many of the lean methods have the intended impact of reducing this waiting time. The use of the cycle time evaluation to evaluate these changes emphasizes the economic benefit of time improvements. The VSM used to assess the expected impact of a change in the production process resulted in savings (lower rejection rates) and to some extent a positive view was due to the fact that there were significant gaps between standardized work and real work; this gap meant that workers did not strictly adhere to assembly standards.

In the case study [9]the findings of this study enabled us to characterize the current state of the fish and shellfish processing, using the fish bone diagram and VSM of the process, and identify the tasks that do not add value to the process, which account for 37.7

percent of the processing time and correspond to process inventories, excess movements due to deficiencies in plant distribution, and bottlenecks that begin during the filleting process. Following that, the waste, or MUDAS, was discovered using techniques such as Pareto diagrams and cause-and-effect diagrams, as well as an examination of the process's primary indicators, such as complaints and returns. Finally, improvement ideas were developed using lean tools, which can enhance plant organization and process speeds. By changing the layout of the process. Which resulted in decreasing the movement by 40 percent and time by 54 percent per unit. Thus have an impact on the company's profit margins and the reduction of product complaints and returns.

In the case study[13]the goal of this article is to investigate the extent to which LM is being implemented in Malaysian automotive manufacturing enterprises, by using survey to collect the data from Federation of Malaysian Manufacturers (FMM) and SME Corp Malaysia directories. According to the findings, the majority of respondent enterprises have implemented lean manufacturing systems to some degree. Cluster analysis is used to group respondent organizations based on the level of lean manufacturing implementation or their status in one of five lean manufacturing practice categories. The majority of respondent firms are categorized as in-transition to lean since their mean values for each of the five characteristics are modest. Because the primary challenges for these organizations are a lack of actual grasp of the lean manufacturing idea and employees' attitudes, they should be made aware of and comprehend the lean concept and goal.

In the case study [18]implementing lean manufacturing using Kanban. The case study carried out qualitative research design based on a Malaysian manufacturing company that deployed Kanban as a part of a lean manufacturing system. Data was collected by meetings, observations and interviews with mangers in the company. The Kanban system implemented in this manufacturing company was found to be adequate due to the numerous benefits such as reduced operational costs, wastes, scraps, and losses, and control of overproduction stocks through flexible work stations. Briefly the new Kanban flow based on workers who attach Kanban cards if the inventory used up, sorting the Kanban cards and loading the bay according to Kanban cards received. The factors that hinder SME companies from implementing the Kanban system are identified as ineffective inventory management, lack of supplier participation, lack of quality improvements and quality control and lack of employee participation and top management commitment. Implication of this study suggest that further research needs to be done on (more small and medium enterprises in Malaysia) so as to have more conclusive findings on Kanban implementation and barriers faced by the SME entrepreneurs. Other than that, the organization must design standard operating procedures for all processes involved in production. Apart from that, the company must build standard operating procedures for all processes engaged in the production line by upgrading the present policy in order to make the production process more efficient in the future, which other manufacturing companies can implement. Where cutting lead time was a constant source of difficulty.

In the case study [7]the research investigates implementing lean manufacturing in a labeling and packing plant. The problem facing the facility is the speed in the digital department processes aren't synchronized with each other, leading to unnecessary

inventory. In addition to time wasted in waiting for goods and proper tools to arrive. Examining the problem scenario, it became clear that the problem might be solved by implementing a lean manufacturing method. Keeping that in mind, VSM was used to illustrate the problem's dimensions, and bottlenecks and their impacts on associated processes were noted. The study's findings may be relevant to businesses operating in the same sector. Die cutting is a key procedure in the printing and packaging sector. It has been discovered that waiting for material flow causes the plant under study to waste a large amount of time. Die-cutting stage inventory holding and setup times are improved by 9.6 % with the use of layout in conjunction with VSM. Furthermore, a 15% increase in the equipment's efficiency has been achieved. The sorting procedure in the finishing area, which is heavily reliant on the product quality, was another process that was emphasized in the mapping. By giving the products a different colored sticker, this has been made better. This colorful sticker contains details about the step a product has generated as well as the quality of the product. Lastly, there was an unnecessary motion in the last stage of a non-value-adding operation; to address this, A new floor flow configuration was implemented, completely doing away with the rubber process and minimizing motion inside the process. The procedure was able to handle 13.04% more goods than previously because to this updated finishing arrangement. Thus, it can be concluded that lean manufacturing powered by VSM has enhanced the process to be more flexible and simplified, helping the company become more competitive and reducing lead times even further.

In the case study [4]the research aims to study the time taken to manufacture connecting rods in a facility. The researcher recorded each machining process by 5 iterations, then state the average, minimum and maximum time taken for each process and plotting the results on graphs. Moreover, a fishbone diagram was constructed to identify the delay time per each process. After implementing the lean concept and an alternative layout for the machine floor, new graphs were plotted by comparing them to the old graphs, the material handling and operational time is reduced. In conclusion, by fixing the manufacturing process's issues, the component's manufacturing time was shortened. The best arrangement for manufacturing connecting rods was identified. Many objectives were accomplished by incorporating lean manufacturing concepts into different processes and slightly altering the unit layouts' path flow, including optimum utilization of machine floor, decrease in motion waste and increase in productivity due to reduction of wasted time.

2.7 Motivation

This is a vital opportunity for an undergraduate student to apply the lean philosophy in a real-life scenario. This is a great step for what was learned in the past 3 years in the university. In addition to having a chance to work in the top water meter facility in Egypt and benefit from the knowledge of the engineers and workers there. The aim of the research is to increase the productivity of the water meter in the facility. Clearly, this will have a massive impact on my knowledge as an undergraduate and it will be a boost for

my career, to apply for an internship abroad. Moreover, lean manufacturing has a high demand in the labor market, because it can be applied in every facility, regardless how big or small it is. Finally, This is will be a great head start for an ungraduate engineer.

Chapter 3

Methodology

This study investigates applying lean manufacturing to improve the performance of a water meter facility. The theory of lean manufacturing should be covered at the beginning of the project, to identify all the lean tools and which one would fit this case. Therefore, a thorough literature review is conducted on the lean philosophy, lean tools and case studies in various industries. The problem of the facility should be identified, along with literature review to collect the data, givens and knowledge from the researches in the industry. After identifying the problem, which is increase the productivity of units per hour in the machining process. Two value stream maps are implemented. The first one shows an overview of all of the processes used in manufacturing the product (water meter) in the facility, the second one focuses on the machining processes of the product. After gathering all the information needed and designing the flow of the product, the wastes in the facility should be identified and eliminated or reduced using the lean tools such as 5S methodology, which were discussed thoroughly in the literature review. Then implement the lean tool which fits the processes. Finally , evaluate the results of the implemented solutions for the facility and compare it with old and new VSM.

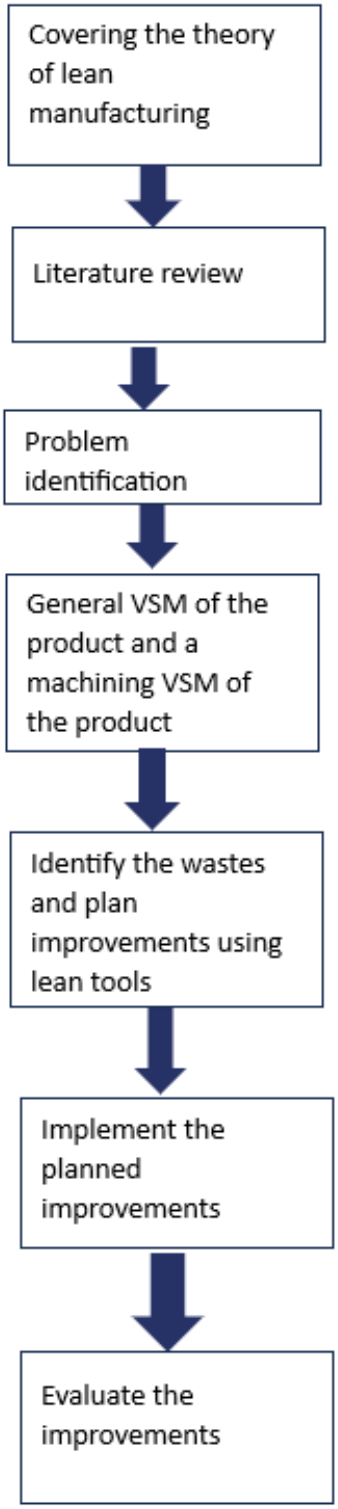


Figure 3.1: Flowchart

Chapter 4

Implementation

4.1 Objective

The objective of this project is to apply lean manufacturing philosophy in Metrotec a water meter facility to improve the productivity of the Zenner mechanical water meter. This project aims to identify and eliminate the wastes that hinder the productivity in the facility. Therefore, a VSM is applied to the whole manufacturing process of the Zenner mechanical water meter. The aim of the VSM is to identify bottlenecks and waste in the manufacturing process. Furthermore, other diagrams that calculates the cycle time, monitors workers movement and operating time are included to give an emphasized picture of the problems that is facing the production department.

4.2 General VSM

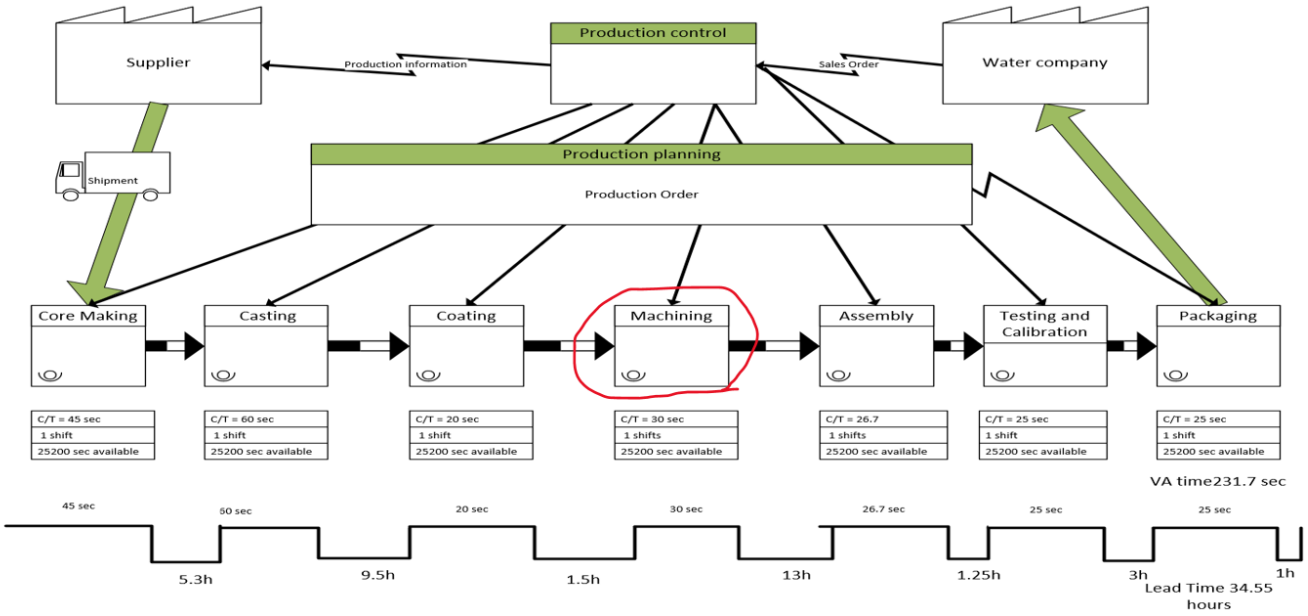


Figure 4.1: General VSM

The General VSM shows the main processes used in manufacturing the Zenner mechanical water meter. The project will focus on reducing the wastes in the machining process.

4.3 Machining VSM

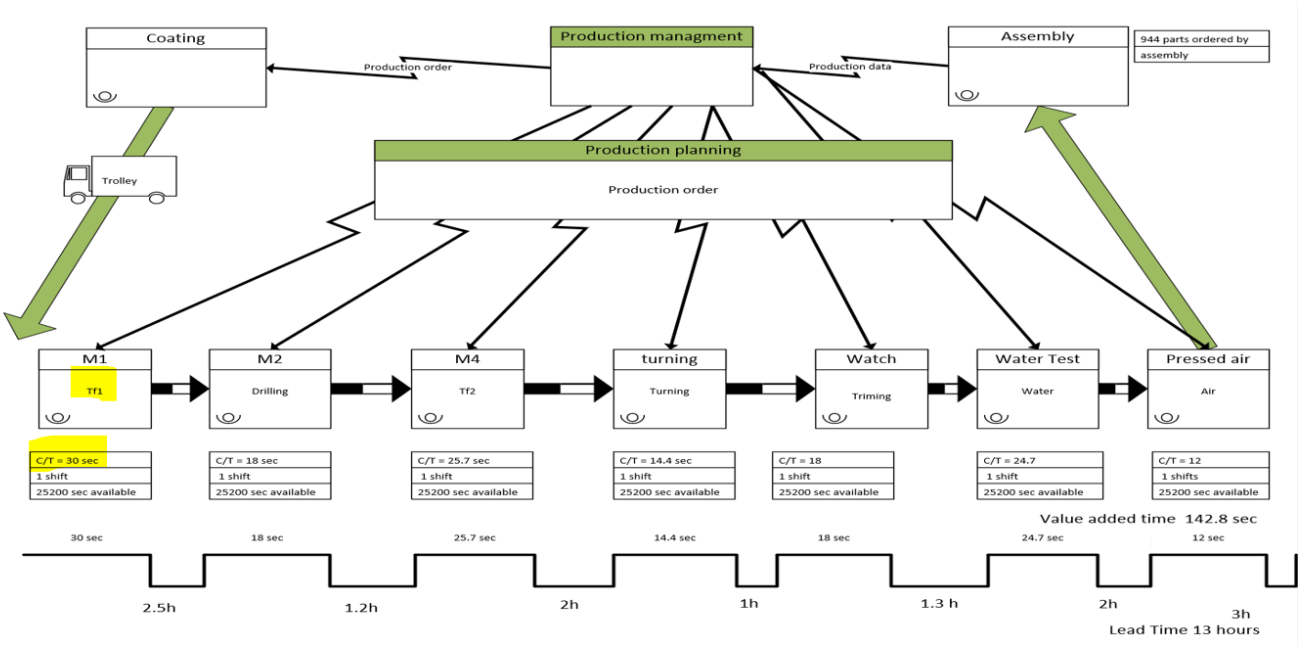


Figure 4.2: Machining VSM

After implementing the general VSM, a second VSM was implemented on the machining process, as there are many wastes in this process, according to the production manager, who wanted to focus on eliminating waste and increasing efficiency in the machining department. Therefore, a second VSM was developed to monitor the waste in the machining department. The Machining VSM which focuses on every machining process. Therefore, it can be inferred that there is a bottleneck in the first operation (Transfer 1 machine)

4.4 Transfer 1 machine and the Zenner



Figure 4.3: Transfer 1 machine

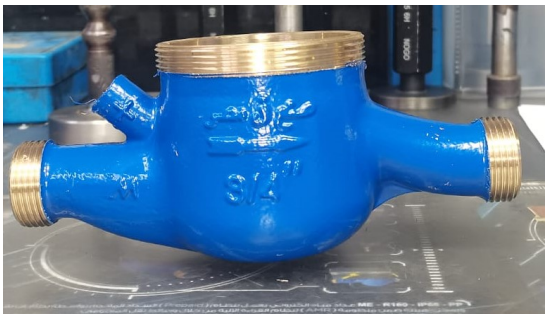


Figure 4.4: Zenner mechanical water meter

Figure 4.5

The worker feeds the transfer 1 machine with a unit and it does 6 operations automatically which are 3 threading and 3 turning. These pictures demonstrate the transfer 1 (TF1) which is the bottleneck process and the other picture is for the Zenner mechanical water meter the unit used.

4.5 Cycle time and Efficiency

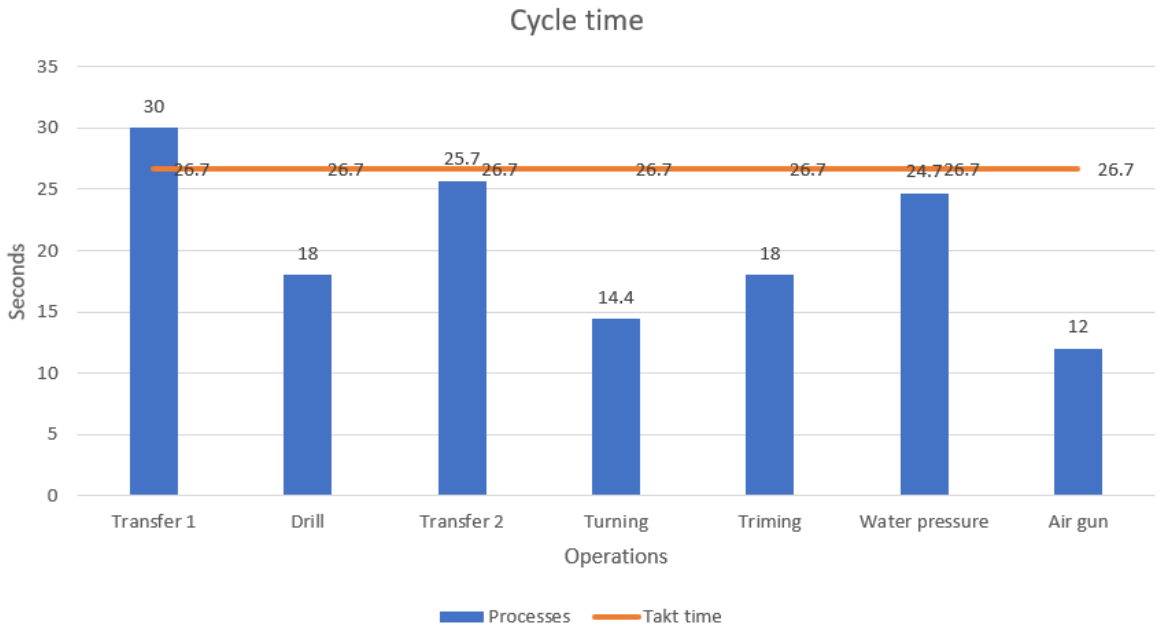


Figure 4.6: Cycle time

This diagram represents the cycle time for every operation and the takt time, which is the time required for unit from the assembly department. After implementing the second VSM and Cycle time graph. As shown in the previous figures. It can be inferred that there is a bottleneck in the first machining process (Transfer machine 1). The aim of this machine is to do turning and threading to the water meter. The cycle time of the assembly department is 26.7 seconds **Takt time** and its productivity 135 unit/h ,while the transfer machine 1 cycle time is 30 seconds and its productivity is 120 unit/h. Therefore, the C/T of the TF1 machine should be reduced to adhere with the takt time of the Assembly department. Another information was inferred from the above figures is that there are processes that has a very low cycle time, such as turning and air gun. The workers in these processes stays idle after finishing their work. Therefore, they could be utilized in other operations. To identify the reason of why the transfer machine 1 cycle time is 30 seconds, a detailed process observation of the TF1 was carried out. The process observation was done on a period of 7 hours in different days and time. Each observation was done on a period equivalent to one hour. The observations took place on different timings of the shift (e.g. Beginning, after break, before the end of the shift). The time study technique will be used is work sampling. Which is observing the operator at normal conditions of work, with considering time allowances.

4.6 Observation of the transfer 1 machine

Observation of the process

Video Ref	3
Date	2/11/2023
Engineer	Abdelraheem Zekry
Study ref	
Part #s	Transfer 1
Shift	Morning
Starts with	11:50
Ends with	12:36

Step#	Description	Time Min/s	Qty	Op	non op.	1=VA 0=NVA	Notes/equipment used
1		1:20	4	○	⇨	1	
2	Worker stopped working	2:00		○	⇨	0	
3		1:15	4	○	⇨	1	
4	Worker is flipping the trolley	0:20		○	⇨	0	
5		10:24	35	○	⇨	1	
6	Bringing new trolley	12:00	0	○	⇨	0	Should take 5 mins max
7		17:40	56	○	⇨	1	Finished all trolleys
8				○	⇨		
9				○	⇨		
10				○	⇨		
11				○	⇨		
12				○	⇨		
13				○	⇨		
14				○	⇨		
15				○	⇨		

VA	**4**
NVA	**3**
% VA	**57%**

99 parts

Figure 4.7: Observation of transfer 1

An observation chart was conducted to observe the worker on transfer machine 1 and analyze his work. This is a sample of one observation chart, 6 more were done. The time taken for the TF1 to finish 1 unit is **19 seconds**. The time study technique used in this research is work sampling. For every observation of process done, the number of units produced are calculated for every part per observation.

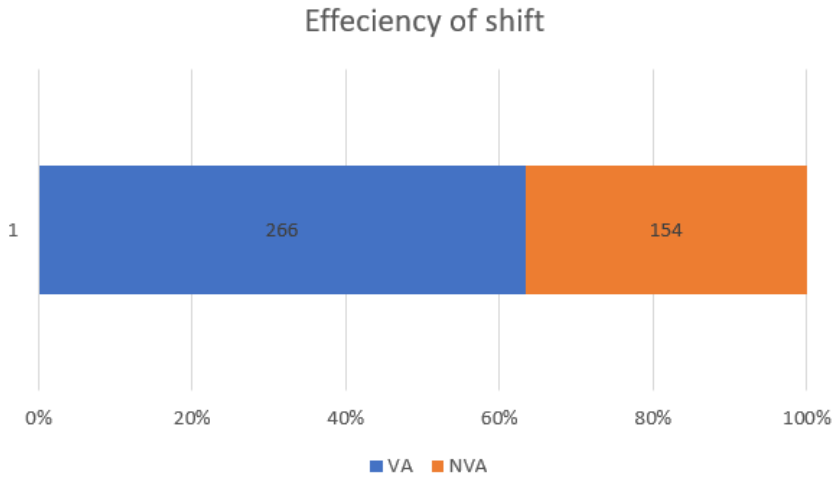


Figure 4.8: Efficiency of shift

This bar graph shows the efficiency of the shift by dividing it into a VA or NVA activities. The total time of the shift is 420 minutes. The VA activities contributes by 2/3 of the shift (266 min). While the NVA contributes by 1/3 of the shift (154 min) time as as shown in the figure.

4.6.1 Value Added (VA)

The VA time is the time taken for the machine to do the machining process of the part and feeding the machine with parts. Which is 19 seconds per part.

4.6.2 Non Value Added (NVA) necessary wastes

NVA necessary wastes is divided into four parts.

- First, the time needed for chip removing and cleaning the machine, which is 30 minutes for every shift (Waiting Waste)
- Second, inspecting parts in the quality department as they might be defected due to wear of tool, it mainly occurs at the beginning of the shift or after changing the cutting tool to make sure that there is no tolerance. This usually takes 8 minutes per shift (Defects waste)
- Third, changing the cutting tool due to wear of the tool and this operation takes 26 minutes, because the worker sends the tool to the lathe department and wait until it is sharpened (Waiting waste)
- Fourth, Flipping the trolley which carries the water meters. It takes 20 seconds to flip the trolley and the worker finishes 6 trolleys per shift. The total time is 6x20sec= 120 sec or 2 minutes. Fifth, bringing new trolley which takes 1 minute for every trolley, so it is 6 minutes for the 6 trolleys (Motion waste)

4.6.3 Non Value Added (NVA) Pure unnecessary wastes

The wastes could be classified as:

- Taking a longer time to bring the trolley from the painting station to store it in the trolley station (Motion Waste)

- Chatting between workers and shutting down the machine (waiting waste)

- Workers may waste up to half an hour to operate the machine at the beginning of the shift (Waiting waste)

- Workers waste a lot of time to come from the break (Waiting waste)

4.7 Spaghetti diagram

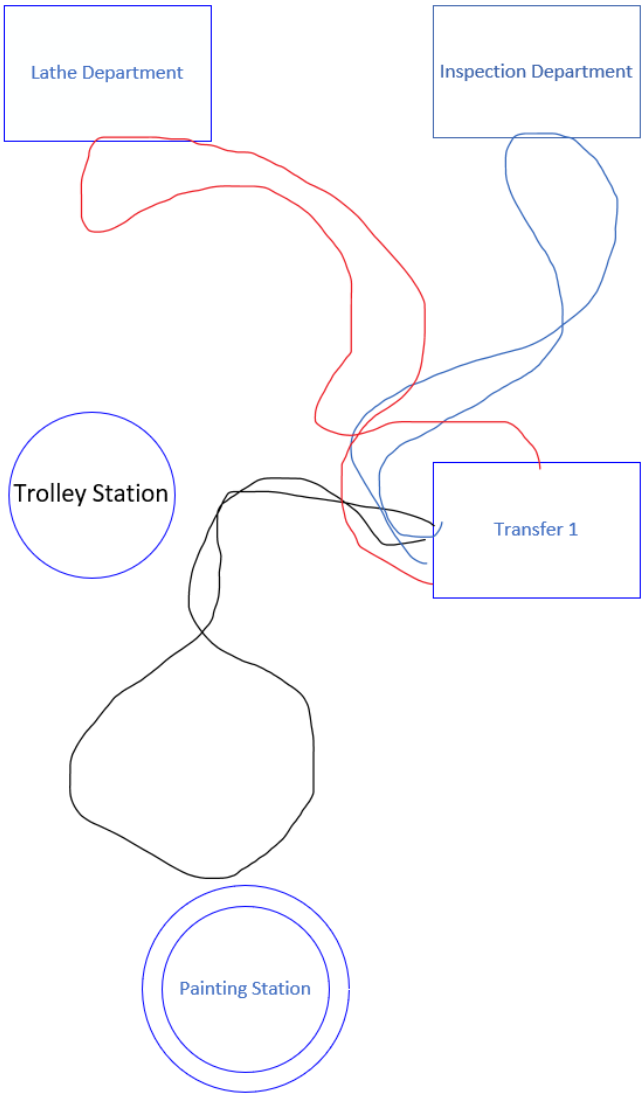


Figure 4.9: Spaghetti diagram

The Spaghetti diagram is used to trace the motion of the worker in a process. This diagram shows the motion of the worker on transfer 1 machine. The workers motion begins with operating the machine then going to the painting station to bring trolleys to store them in the trolley station. The worker goes to the inspection department, if he can see any defects in the product or check for the measurements . The worker goes to the lathe department to change the tool if it worn out or broke down. The workers motion

could be by eliminated to the painting station and Lathe department. By assigning the work to another worker that brings the trolleys from painting to trolley station

Key: Red lines represents the motion of the worker to the lathe department. Black lines represent he motion of the worker to bring trolleys from painting to trolley station. blue lines represents the motion of the worker to the inspection department.

4.8 Wastes Identification and Targeted Improvements

After analyzing the situation of the transfer 1 machine by using lean approach. The suggested improvements can be mentioned as follows. First, (NVA pure unnecessary wastes) must be eliminated or reduced, as it will save around 82 minutes per shift. Second, the air gun operator becomes idle after finishing his task as his cycle time is 12 seconds only. Therefore, he could be assigned cleaning the transfer machine from chips (this is a NVA necessary wastes that takes 30 minutes every shift) and bringing trolleys from painting station to trolley station (this is a NVA necessary wastes which takes around 30 minutes every shift).This will result in a better utilization of workers. Third, there should be a spare tool ready for operation in the lathe department. This could decrease the tool change time, which takes minimum of 26 minutes. The aim of this improvements is to decrease the C/T of TF1 machine and make its production not less than 135 unit/h, so it can adhere with the assembly department. The planned improvements are briefly mentioned in the following points.

- Eliminate and reduce the NVA pure unnecessary wastes, this will save 82 minutes per shift (waiting waste)

- Assign the air gun operator the chip removing (waiting waste) and trolley transportation to the trolley station (Motion waste) tasks. This will lead to better utilization of workers

- Ready spare tool in the lathe department

4.9 Utilization of workers

4.9.1 Planned utilization of machines

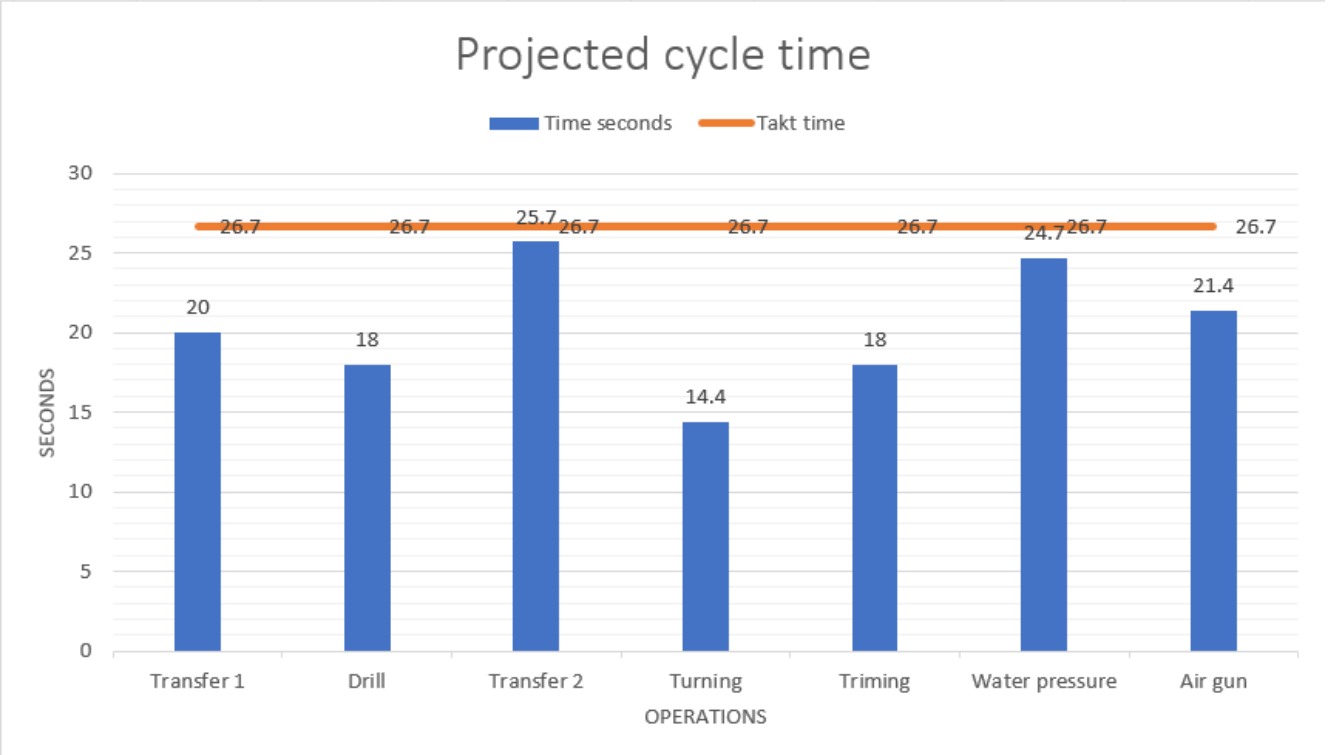


Figure 4.10: Projected cycle

This is the planned cycle time for the transfer 1 and air gun operations after modifications using the line balancing technique. In this diagram the worker on the air gun operation will be assigned the operations of cleaning the chips in the machine and bringing trolleys from the painting station to the trolley station. This will result in decreasing the cycle time of the Transfer 1 operation from 30 sec to 20 sec and increasing the cycle time of the air gun from 12 sec to 21.4 sec. Therefore this adheres with the takt time.

4.9.2 Standard work combination table

4.9.3 Planned utilization of machines

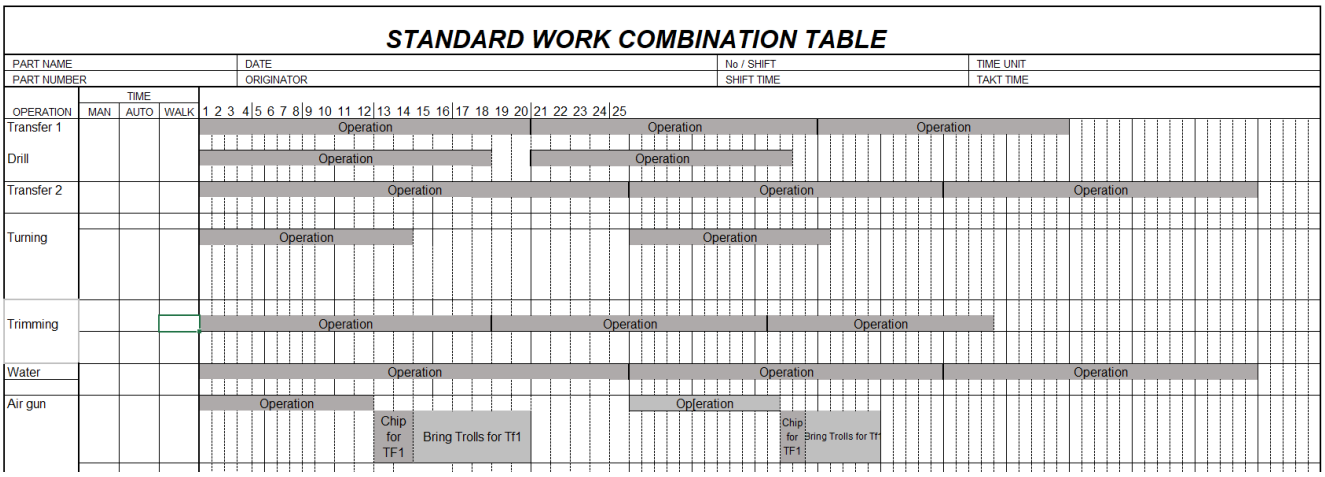


Figure 4.11: Standard work combination table

The standard work combination table is presents every C/T in the machining processes. Here it is used to show that if the modifications could be applied by presenting the new cycle time with assigning the air gun operator the chip removing and bringing trolleys to the trolley station tasks. If the C/T of the operation in the air gun worker is 12 seconds he can also be assigned removing chips and storing trolleys tasks which will take around 8 seconds.

4.10 5S methodology

5S is a methodology used to create more organised and productive workspace. Therefore this project will focus on applying 5s standards in the lathe department, to create a well organized space for 3 products tools, which are the Zenner mechanical water meter, Prepaid water meter and the Zenner cover. As the worker takes more than 26 minutes to change the tool in the transfer 1 machine. The aim is to create a well organised lathe department, as this will reduce the NVA necessary wastes. Therefore it will reduce the cycle time of the transfer 1 machine. The problems that causes time waste in the lathe department that it is unorganized, the tools are not sorted according to their type or size, there is no certain place for the defected or working tools, there is no spare tool ready for operation and the department is stuffed with unnecessary parts.

4.10.1 Sort

The first step in the 5S methodology is to implement sorting in the lathe department. The sorting step separates the needed items in the workspace from the not needed items

from the workspace. Therefore, the unneeded items from the workspace in the lathe department are moved away. Then create cards for tools written on the product type and size using 2 colors(Green and Red). The green color card means that the tool is ready for use and has no defects. The red card means that the tool has a defect or should be sharpened.

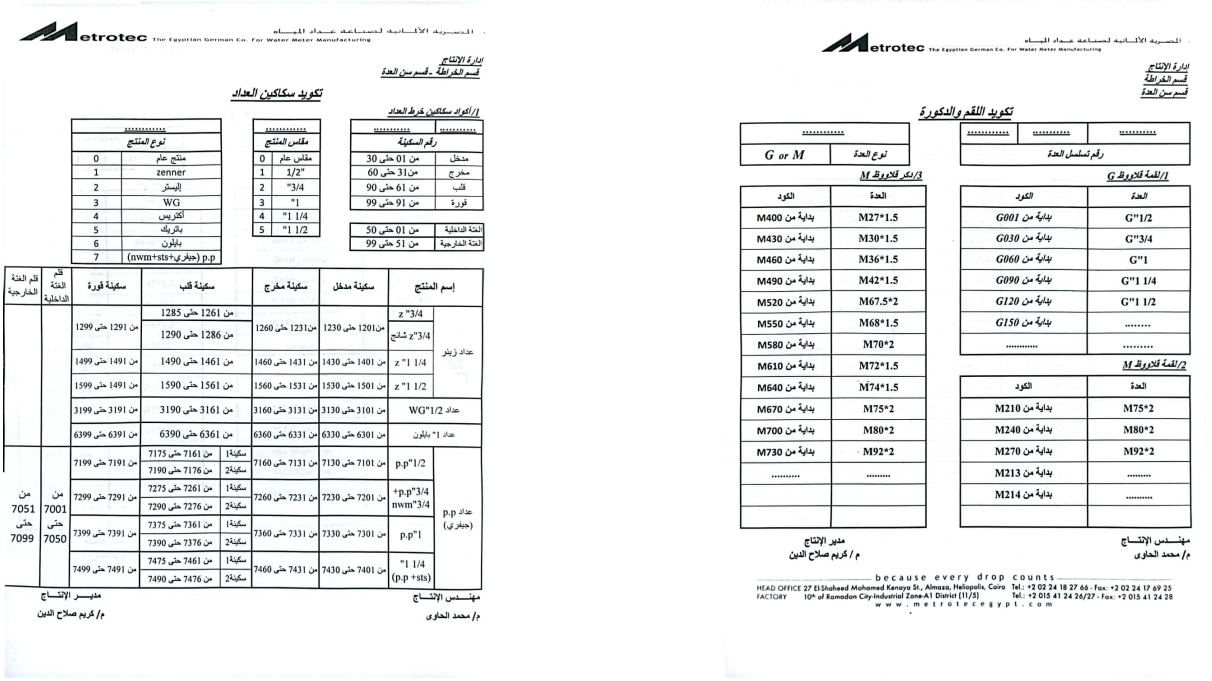


(a) (b)

Figure 4.12: Cards

4.10.2 Straighten

The straighten step means to thoroughly organize the workspace after the sorting step to make the frequently used items is easy to find. Therefore, the green and red cards are sticked on the shelves of the lathe department to give a visual effect for the worker to place the tools according to its condition on the right card. The bottom shelf will have the red cards while the upper shelf will have the green cards. There is a cardboard barrier between each product so the products doesn't get mixed together. Number the tools with a code according to its type and size, then created a sheet with the tools code. This sheet will be hanged on the wall to make it visually easy for the operator to find the code of the tool. This method is applied only on 3 products the zenner mechanical water meter, Prepaid water meter and the Zenner cover.



(a) (b)

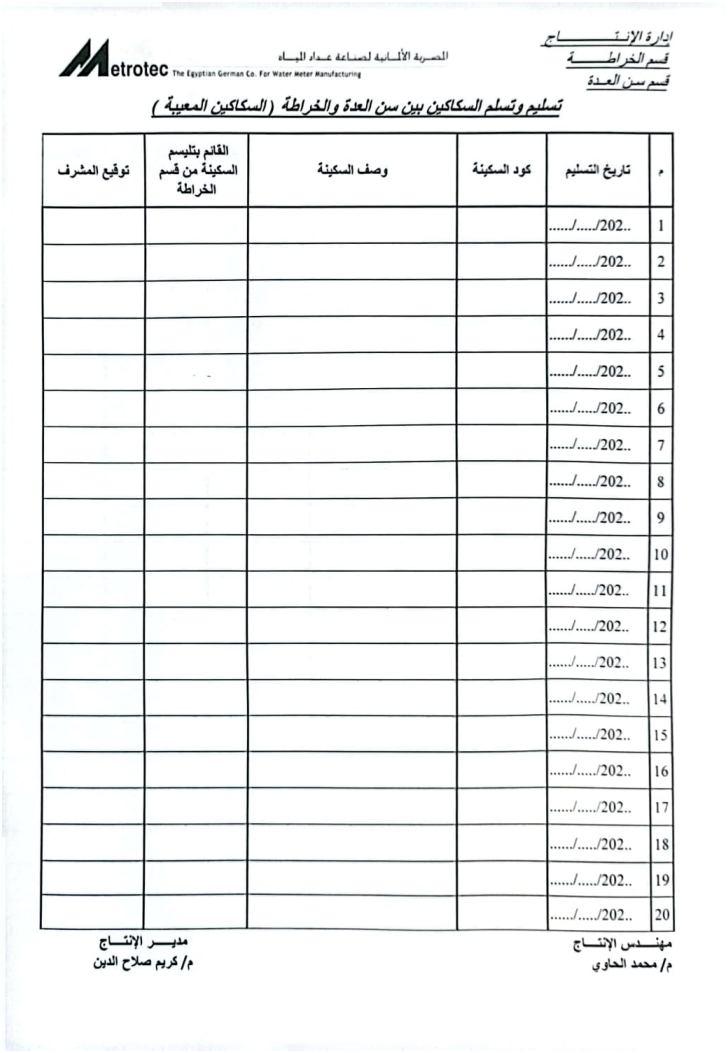
Figure 4.13: Tool codes

4.10.3 Shine

In the shine step elevates the work area by thoroughly cleaning and inspecting tools. This is done by cleaning the shelves from dust, clean and wipe the tools to make the cracks visible. Because the cracks could easily lead to failure of tools.

4.10.4 Standardize

In this step is the goal to capture the best practices for the 5S methodology. This is done by creating Created a card for every tool change, the card contains sections of date, operator signature, supervisor signature, tool code and failure description. The operator should fill the paper every time he changes the tool.



إدارة الإنتـــــــاج
قسم الخراطـــــة
قسم سن العـدة

المصرية الألمانية لصناعة عداد المياه
Metrotec The Egyptian German Co. For Water Meter Manufacturing

تسليم وتسلم السكاكين بين سن العدة والخراطة (السكاكين المعيبة)

م	تاريخ التسليم	كود السكينة	وصف السكينة	القائم بتليسم السكينة من قسم الخراطة	توقيع المشرف
1	...../...../202..				
2	...../...../202..				
3	...../...../202..				
4	...../...../202..				
5	...../...../202..				
6	...../...../202..				
7	...../...../202..				
8	...../...../202..				
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10	...../...../202..				
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12	...../...../202..				
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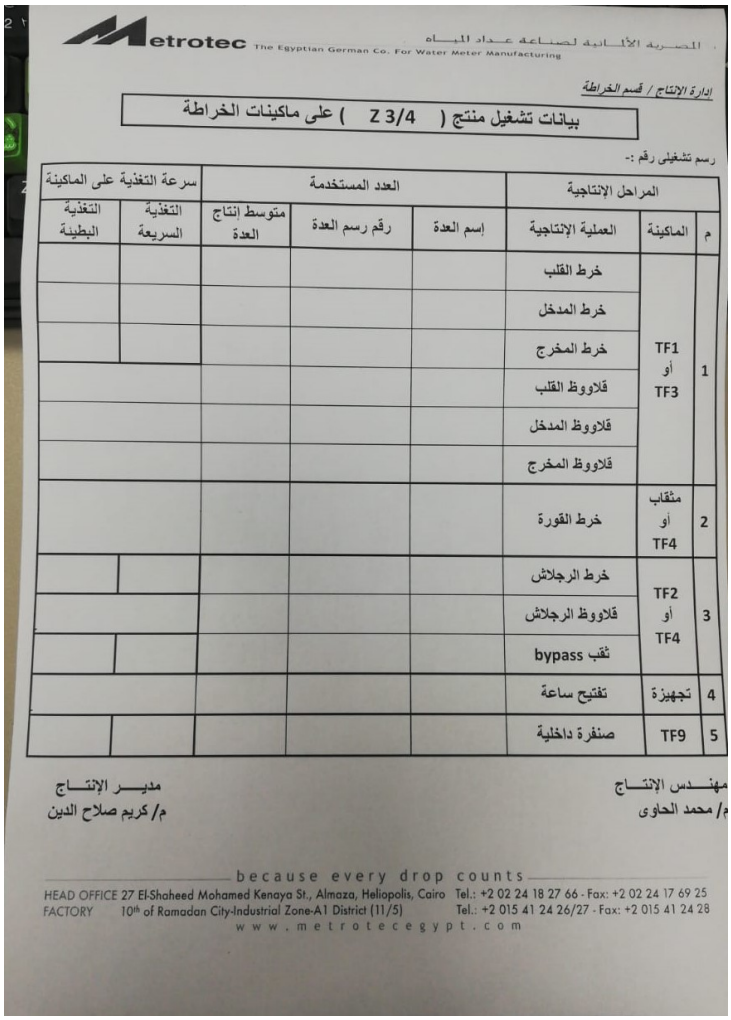
مهنـــدس الإنتـــاج
م/ محمد الحاوي

مديـــر الإنتـــاج
م/ كريم صلاح الدين

Figure 4.14: Signature paper

4.10.5 Sustain

The Sustain step assures that 5S is applied on an ongoing basis. Therefore, the Signature paper is crucial to make sure that the tool change is done according to the 5S steps that has implemented before. In addition to a check sheet could be made for weekly basis to have sections for cleaning, maintenance and organizing the lathe department. Finally, designed a card with the right tool type, feed and pressure for every machine in the lathe department.



Metrotec The Egyptian German Co. For Water Meter Manufacturing

المصرية الألمانية لصناعة عداد المياه

إدارة الإنتاج / قسم الخراطة

بيانات تشغيل منتج (Z 3/4) على ماكينات الخراطة

رسم تشغيلى رقم :-

م	الماكينة	العملية الإنتاجية	إسم العدة	رقم رسم العدة	متوسط انتاج العدة	التغذية السريعة	التغذية البطيئة
			المراحل الإنتاجية	العدد المستخدمة		سرعة التغذية على الماكينة	
1	TF1 أو TF3	خرط القلب					
		خرط المدخل					
		خرط المخرج					
		قلاووظ القلب					
		قلاووظ المدخل					
		قلاووظ المخرج					
2	مثقاب أو TF4	خرط القورة					
3	TF2 أو TF4	خرط الرجلاش					
		قلاووظ الرجلاش					
		ثقب bypass					
4	تجهيزة	تفتيح ساعة					
5	TF9	صنفرة داخلية					

مديـــر الإنتـــاج
م/ كريم صلاح الدين

مهنـــدس الإنتـــاج
م/ محمد الحاوى

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Figure 4.15: Specifications of machines

4.11 Lathe department

This section compares the lathe department before and after the modifications.

4.11.1 Before

This is the lathe department before implementing the modifications



Figure 4.16: The lathe department before modifications



Figure 4.17

Figure 4.18: This is the lathe department before implementing the modifications.

The shelves in the lathe department are dusty. The tools are scattered on the shelves and are not classified and it is not differentiated from the operating tools and non operating tools.

4.11.2 After

This is the Lathe department after the modifications



Figure 4.19



Figure 4.20

Figure 4.21: After modifications



Figure 4.22: After Full modifications

The shelves are clean. There are barriers between every product. It is easy to differentiate between operating and non operating tools by just looking on the card color. The tools code paper is hanged on the wall so the worker can get the right tool easily. The signature paper is on the shelves so the when a tool is changed the reason of failure is written with the date and time.

Chapter 5

Evaluation and Discussion

5.1 Evaluation

This section evaluates the planned improvements and the implemented solutions. Therefore, three runs were made after the modifications and suggestions towards the transfer machine 1. each run lasts around 2 hours. The circumstances of each run is that the worker must have the trolleys near him, the chips should be cleaned without stopping the machine, if the worker wants to have a rest or go to the toilet another worker should cover his absence and a spare tool is ready if a tool wear out. This is done with an assist of another worker to bring the trolleys or clean chips.

5.1.1 Run 1

In the first run the worker didn't leave his place and was working for 2 hours non stop. But the problem is that he finished 260 parts only after 2 hours. This means that every product took around 27.7 seconds. The worker usually takes around 19 seconds to finish one product if he is working non stop. Therefore this lead to 2 discoveries. First, the worker was speeding up the machine to finish his work early, regardless of the excess use of the machine and wear of tools. Second, the facility doesn't have a reference for the right feed or pressure to be used for the machine, however the feed they are using was based on previous attempts without having a refrence.

5.1.2 Run 2

in the second run the operator increased the pressure and the feed of the machine. This resulted in a decrease of the machining time to be 26.3 seconds. This machining time achieves the project aim, which is less than the cycle time of the assembly station. However there could be better results.

5.1.3 Run 3

In the beginning of this run the machining time was around 26 seconds for the first 83 parts. But the production manager adjusted the feed and the pressure for the machine. This reflected on the machining time to be with an average of 23.7 seconds or less than 24 seconds. By this new feed and pressure for the machine, it can be inferred that the new machining time of the machine is 24 seconds in the ideal condition (without time being wasted). Apart from this there was a tool that failed in the middle of the operation. the time taken for the tool change is 11 minutes for disassemble and assemble the new tool and 3 minutes for inspecting the tolerances. this is less than the old method of the tool change which took not less than 26 minutes for lathing the tool.

5.2 Discussion

After implementing the lean tools and evaluating the work conditions by 3 runs. First, it can be inferred that the time taken to finish 1 part is 24 seconds not 19 seconds as the workers used to increase the feed and pressure of the machine to finish their work early. This misuse of the machine decreases the life time of the tools and the machine parts. The new cycle time of transfer 1 is 24 seconds not 30 seconds. Therefore, after the improvements the machine could produce 150 unit/h which exceeds the target of the project which was 135 unit/h. Second, After applying the 5S methodology in the lathe department, this made the tools are well organized and sorted according to their product type and their condition. Moreover, the tool change took much less time than lathing the worn out tool and reinstall it. Third, assigning the air gun worker the chip removing and bringing trolleys to the trolley station tasks. This reduced the NVA necessary wastes by around 65 minutes. Fourth, if the unnecessary wastes were eliminated or reduced it could save around 82 minutes of work per shift. Finally, the TF1 machine could produce up to 150 units per hour. However, remains a problem with the workers as they waste a lot of unnecessary time.

5.3 Final VSM

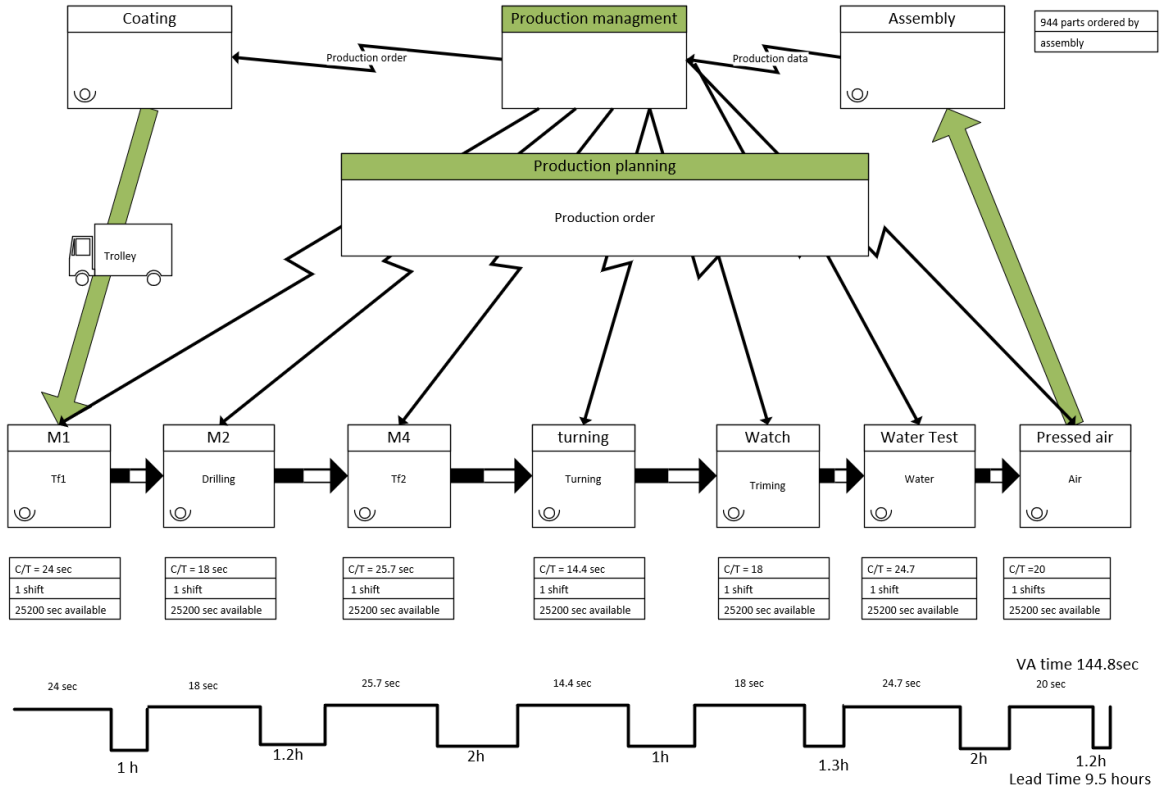


Figure 5.1: Final VSM after modifications

In the VSM the C/T of the TF1 machine is decreased from 30 to 24 seconds due to the planned improvements. The C/T of the pressed air operation is increased from 12 to 20 seconds due to assigning the operator the NVA tasks of the TF1.

Chapter 6

Future recommendations

While this study contributes valuable insights in improving the productivity in a machining process using lean manufacturing tools, several problems should be explored further and solved.

1. There could be more workers hired in the facility especially in the machining department. As there are around 5 workers per shift for 14 machines. Although not all the machines work together in 1 shift, but it is still insufficient amount of workers related to the number of machines to finish work especially in high demands.

2. There could be a financial motivation for workers. Implementing a bonus strategy related to the output produced per unit. This will be a great motivating factor to the employees. Also the workers wages could be increased due to the inflation.

3. The safety standards should be increased in the facility. Wearing safety gloves and goggles for every worker should be mandatory.

4. Maximum efficiency rate of every machine should be calculated(maximum pieces per hour). This can result in finding hidden problems in the production operation.

Chapter 7

Conclusion

In conclusion this project aims to improve the productivity of a water meter facility using the lean manufacturing methods. A thorough literature review was implemented on lean philosophy. The problem was identified by the facility which is improving the productivity in the machining process using lean tools. Data was gathered using VSM of the whole process of the water meter, then it appears a bottleneck in the machining process. A further VSM was conducted on the whole machining process. It showed a bottleneck in the transfer 1 machine . The aim of the project is to decrease the cycle time of the machine and to find out what is hindering the productivity of the machine. An observation of the worker and machine was done at different times of the shift and every operating and non-operating step is recorded. Then several diagrams were conducted using the data gathered from the observation process. After, gathering and analyzing the data it showed out an efficiency of 66% of the shift(VA) and the rest is (NVA pure unnecessary wastes and necessary wastes) . Each type of wastes were identified according to NVA pure unnecessary wastes and necessary wastes. Improvements were planned to reduce or eliminate the NVA activities. Planned improvements were implemented such as assigning the air gun operator NVA necessary tasks and 5S methodology to organize the lathe department. Evaluation was done to the planned improvements by initiating 3 runs with circumstances. The planned improvements led to decrease of the C/T of TF1 machine from 30 to 24 seconds. The new productivity of TF1 is up to 150 unit/h. This exceeds the target of the assembly which is 135 unit/h

Appendix

Appendix A

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